

BAULKHAM HILLS HIGH SCHOOL



MATHEMATICS ADVANCED ASSESSMENT

December 2011

*Time allowed: 50 minutes
plus 5 minutes reading time*

STUDENT NUMBER : _____

TEACHER'S NAME: _____

QUESTION	MARK
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
TOTAL	
PERCENTAGE	

Topics Tested: *Integration and Series*



Advanced Mathematics

December 2011

Time: 50 minutes + 5 minutes reading time

DIRECTIONS

- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.
- Write your teacher's name and your name on the cover sheet provided
- At the end of the exam, staple your answers in order behind the cover sheet provided, and your questions on the back
- Approved Maths aids and calculators may be used

1. Find the primitive function for $3x^2 - \frac{1}{\sqrt{x}}$ 2

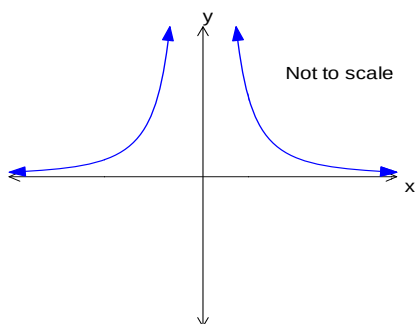
2. Evaluate $\int_1^4 (3x - 5)^2 dx$ 2

3. Given that the limiting sum of $15 + 15x + 15x^2 \dots$ is 45, find x 2

4. The sum of the first 4 terms of an arithmetic progression is 12, and the 15th term is 75. Find the first term and the common difference. 3

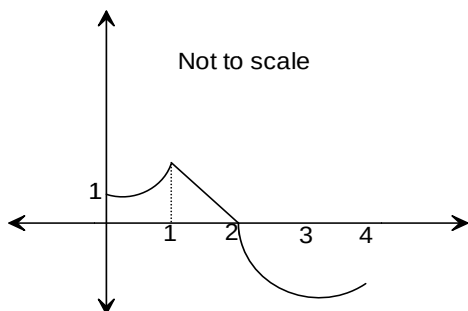
5. The first three terms of an arithmetic series are -2+3+8+....
(i) Find the 60th term. 2
(ii) Hence or otherwise, find the sum of the first 60 terms of the series. 2

6. Given the graph of $y = \frac{1}{x^2}$ below 2

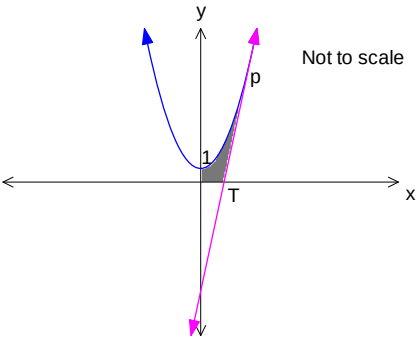
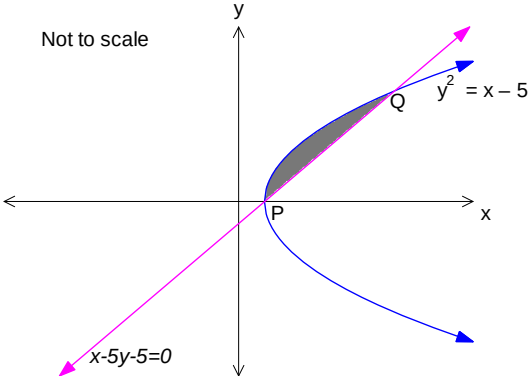


Find the area bounded by this graph, the x -axis and ordinates $x = 1$ and $x = 4$.

7. Given the graph of $y = f(x)$ below, find the approximate area bounded by the curve, the x -axis and the lines $x = 0, x = 4$, using Simpson's Rule with five function values. 3



x	0	1	2	3	4
$f(x)$	1	2	0	-4	-3

8.	The diagram shows the graph of the parabola $y = x^2 + 1$. A tangent is drawn to the parabola at the point P(3, 10)  (i) Show that the equation of the tangent is $y = 6x - 8$. 2 (ii) T is the x-intercept of the tangent. Find the coordinates of T. 1 (iii) Find the area of the shaded region. 3	
9.	 The diagram shows the graph of the curves $f(x): y^2 = x - 5$ and the line $g(x): x - 5y - 5 = 0$. (i) Show that the points of intersection between curves $f(x)$ and $g(x)$ are P(5, 0) and Q(30, 5). 2 (ii) Find the volume of the solid generated by rotating the shaded region around the x-axis. 3	
10.	Bianca would like to save \$60000 for a deposit on her first home. She decided to deposit at the start of each month her net monthly salary of \$3000 in a bank account that pays interest at a rate of 6% per annum compounded monthly. Bianca intends to withdraw \$$E$ at the end of each month from this account for living expenses, immediately after the interest has been paid. Let A_n be the amount in the account after the n-th withdrawal i) Show that the amount of the money in the account following the second withdrawal of \$$E$ is given by 2 $A_2 = \$3000(1.005^2 + 1.005) - E(1.005 + 1)$ ii) Show that the A_n is given by 2 $A_n = (3015 - E) \frac{1.005^n - 1}{0.005}$ iii) Calculate the value of E if Bianca is to reach her goal after 4 years. 2	
~ END OF EXAM ~		

BS Q. 1-7 → 18 marks
 PS Q. 8, 9, 10 → 17 marks
 Unit Ass. task Dec. 2011

out of (35)

Answers

$$\textcircled{1} \int 3x^2 - \frac{1}{\sqrt{x}} dx = x^3 - \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= x^3 - 2x^{\frac{1}{2}} + C$$

$\downarrow$ $\downarrow$ $\nearrow$ if missing
 $\textcircled{1}$ $\textcircled{1}$ $\textcircled{-1}$

$$\textcircled{2} \int_1^4 (3x-5)^2 dx = \left[\frac{(3x-5)^3}{9} \right]_1^4 \textcircled{1}$$

$$= \left[\frac{7^3}{9} - \frac{(-2)^3}{9} \right] = \boxed{39} \textcircled{1}$$

$$\textcircled{3} 15 + 15x + 15x^2 + \dots = 45$$

$$\therefore 45 = \frac{15}{1-x} \textcircled{1}$$

$$\therefore \boxed{x = \frac{2}{3}} \textcircled{1}$$

$$\textcircled{4} S_4 = \frac{4}{2}(2a + 3d)$$

$$\therefore 12 = 2(2a + 3d)$$

$$T_{15} = a + 14d$$

$$\therefore 75 = a + 14d$$

$$\therefore \boxed{d = \frac{144}{25}} \textcircled{1} \quad \boxed{a = -\frac{141}{25}} \textcircled{1}$$

$$\textcircled{5} -2 + 3 + 8 + \dots \quad \text{A.P.}$$

$$\textcircled{1} a = -2 \quad d = 5$$

$$\text{i) } T_{60} = a + 59d = -2 + 59 \times 5$$

$$= 293 \textcircled{1}$$

$$\text{ii) } S_{60} = \frac{60}{2}(-2 + 293) \textcircled{1}$$

$$= 8730 \textcircled{1}$$

$$\textcircled{6} y = \frac{1}{x^2}$$

$$A = \int_1^4 \frac{1}{x^2} dx = \left[\frac{x^{-1}}{-1} \right]_1^4 \textcircled{1}$$

$$= \left[-\frac{1}{4} - \left(-\frac{1}{1} \right) \right] = \frac{3}{4} \textcircled{1}$$

$$\textcircled{7} A \doteq \frac{h}{3} [1 + 4 \times 2 + 0] + \left| \frac{1}{3} [0 + 4 - 4] \right|$$

$$h=1 \textcircled{1} \quad \quad \quad + -3$$

$$\therefore A = \frac{1}{3} [1 + 8 + 0] + \left| \frac{1}{3} [0 + -16 - 3] \right|$$

$$= 3 + \left| -\frac{19}{3} \right| = \frac{28}{3} \textcircled{1}$$

if no abs. value for negative

$$\therefore A \doteq \frac{1}{3} [1 + 4(2 + -4) + 2 \times 0 + -3]$$

$$= -\frac{10}{3} \quad \therefore \boxed{\frac{2}{3}} \textcircled{2}$$

8) i) $y = x^2 + 1$ (6)

$$\frac{dy}{dx} = 2x$$

$$m = 2 \times 3 = 6 \text{ (1)}$$

$\therefore$ tangent

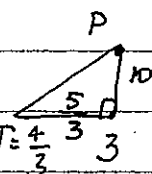
at $P(3, 10) \therefore y - 10 = 6(x - 3)$ (1)

$$\therefore y = 6x - 18 + 10$$

$$y = 6x - 8 \therefore \text{shown}$$

ii) $y_T = 0 \therefore 0 = 6x - 8 \therefore x = \frac{4}{3}$

$$\therefore T\left(\frac{4}{3}, 0\right) \text{ (1)}$$

iii) Area = $\int_0^3 x^2 + 1 dx -$  (1)

$$= \left[\frac{x^3}{3} + x \right]_0^3 - \frac{1}{2} \times \frac{5}{3} \times 10$$

$$= (9 + 3) - 0 - 5 \times \frac{5}{3} = \frac{11}{3} \text{ (1)}$$

9) i) $y^2 = x - 5 \therefore x = y^2 + 5$ (5)

$$x - 5y - 5 = 0 \therefore x = 5y + 5$$

$$\therefore y^2 + 5 = 5y + 5$$

$$y^2 - 5y = 0 \text{ (1)}$$

$$y(y - 5) = 0$$

$$\therefore y = 0 \text{ or } y = 5 \text{ (1)}$$

$$\therefore x = 5 \times 0 + 5 \quad x = 5 \times 5 + 5$$

$$\therefore P(5, 0), Q(30, 5)$$

ii) $V = \pi \int_5^{30} f(x) dx - \pi \int_5^{30} g^2(x) dx$

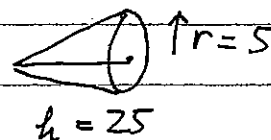
$$\therefore V = \pi \int_5^{30} (x - 5) dx - \pi \int_5^{30} \frac{(x - 5)^2}{5^2} dx \text{ (1)}$$

$$= \pi \left[\frac{x^2}{2} - 5x \right]_5^{30} - \pi \left[\frac{(x - 5)^3}{25 \times 3} \right]_5^{30} \text{ (1)}$$

$$= \frac{625\pi}{6} \text{ or } 327.25 \text{ (1)}$$

(OR)

$$V = \pi \int_5^{30} (x - 5) dx \text{ (1)}$$



$$= \pi \left[\frac{x^2}{2} - 5x \right]_5^{30} - \frac{1}{3} \times 25 \times 5\pi \text{ (1)}$$

$$= \frac{625}{2} \pi - \frac{625}{3} \pi = \frac{625\pi}{6} \text{ (1)}$$

Question 10 (6)

$$i) A_1 = 3000 \left(1 + \frac{6 \div 12}{100}\right) - E = 3000 \times 1.005 - E \quad (1)$$

$$\begin{aligned} A_2 &= (A_1 + 3000) \times 1.005 - E \\ &= (3000 \times 1.005 - E + 3000) \times 1.005 - E \quad (1) \\ &= 3000 \times 1.005^2 - E \times 1.005 + 3000 \times 1.005 - E \\ &= 3000 (1.005^2 + 1.005) - E (1.005 + 1) \text{ : shown} \end{aligned}$$

$$ii) A_n = 3000 \underbrace{\left(1.005^n + 1.005^{n-1} + \dots + 1.005\right)}_{\substack{\text{G.P } a=1.005 \quad r=1.005 \\ n=n}} - E \underbrace{\left(1.005^{n-1} + \dots + 1\right)}_{\substack{\text{G.P } a=1 \\ r=1.005 \quad n=n}} \quad (1)$$

$$A_n = 3000 \times 1.005 \frac{1.005^n - 1}{1.005 - 1} - E \times 1 \times \frac{1.005^n - 1}{1.005 - 1} \quad (1)$$

$$\therefore A_n = \frac{1.005^n - 1}{0.005} \left[3000 \times 1.005 - E \right]$$

$$\therefore A_n = \frac{1.005^n - 1}{0.005} [3015 - E] \quad \therefore \text{shown}$$

$$iii) A_n = 60000 \quad n = 4 \times 12 = 48$$

$$\therefore 60000 = \frac{1.005^{48} - 1}{0.005} [3015 - E] \quad (1)$$

$$\frac{60000 \times 0.005}{1.005^{48} - 1} = 3015 = -E \quad (1)$$

$$\therefore E = \$1905.898 = \$1905.90$$

BS $\rightarrow 1-7 \rightarrow 18 \text{ marks}$

PS $\text{Q. 8, 9, 10} \rightarrow 17 \text{ marks}$

